Applied AI

What is AI and building our own model.

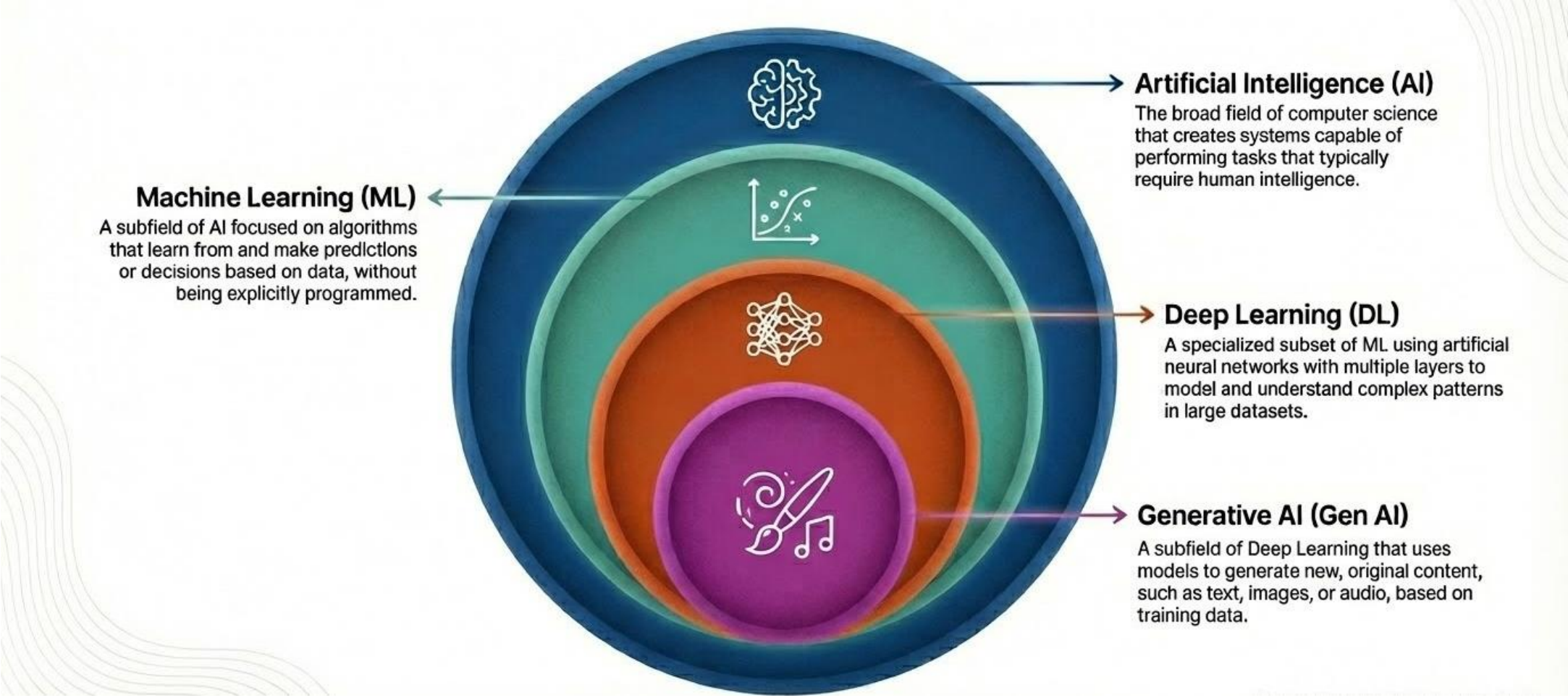
Goals of this lecture…

- Understand how key terms of AI, Machine Learning, Deep Learning and Generative AI fit together.
- Understand the feedforward neural network model.
- Work though a simple 2-layer feedforward neural network model.

What is AI?

- **Artificial Intelligence (AI)** is a broad field of computer science that enables computers to perform tasks that typically require human intelligence.
- These tasks include:
  - **Learning**: Ability to improve performance from experience and data.
  - **Reasoning**: Drawing conclusions and making decisions based on information.
  - **Perception**: Interpreting sensory information, such as images and speech.
  - **Problem Solving**: Finding solutions to complex problems.
- AI encompasses various subfields, including Machine Learning and Deep Learning.

Some of the AI layers



Generated using Nano Bananas [3]

Traditional Approach

As outlined in [2], the traditional approach to AI was to:

"*Ask human experts how they do it, write it down as IF-THEN rules, explicitly program these rules into the computer*"

An example of such AI systems is MYCIN (1970s):

- One of the most famous early AI expert systems.
- Designed at Stanford to diagnose bacterial infections and recommend antibiotics.
- It used hundreds of IF–THEN rules.

Polanyi's Paradox

"We know more than we can tell." - **Michael Polanyi**

EXPLICIT KNOWLEDGE

- Articulable rules & facts
- Step-by-step procedures
- Easy to codify & program

TACIT KNOWLEDGE

- Intuitive understanding
- Muscle memory & skills
- Pattern recognition
- Difficult to articulate & program

The Paradox's Implication for AI: Traditional, rule-based AI struggles with tasks requiring tacit knowledge, highlighting the need for Machine Learning to learn from examples and data, not just explicit instructions.

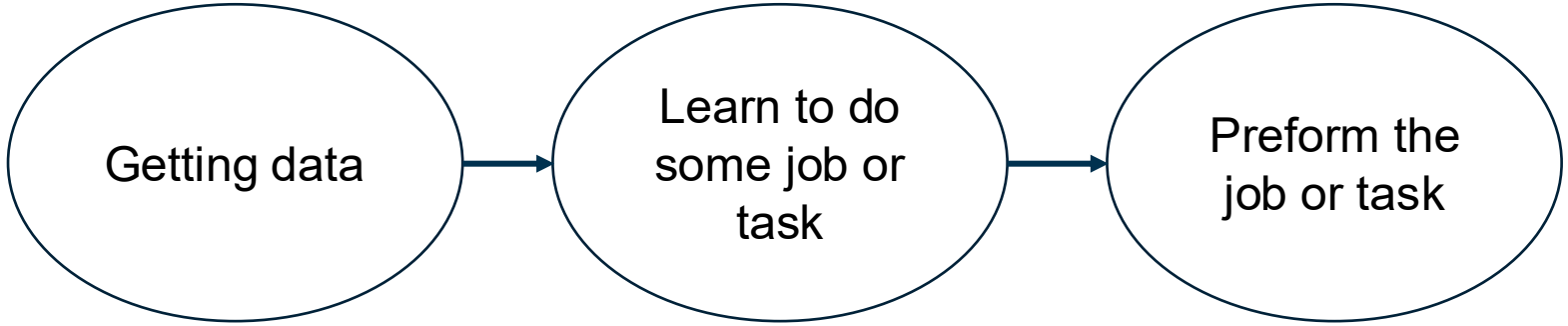
Generated using Nano Bananas [3]

What is machine learning?

A common definition of machine learning is that:

"Machine learning is the study of getting computers to learn without being explicitly programmed"

This is a little vague but in practice machine learning is about:



Tom Mitchell's Definition of Machine Learning & Performance Measures

"A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E." - Tom Mitchell

Deconstructing the Definition

Experience (E): The data the system learns from (e.g., a database of images, historical prices).

Task (T): The specific problem the system is trying to solve (e.g., classifying an image, predicting a value).

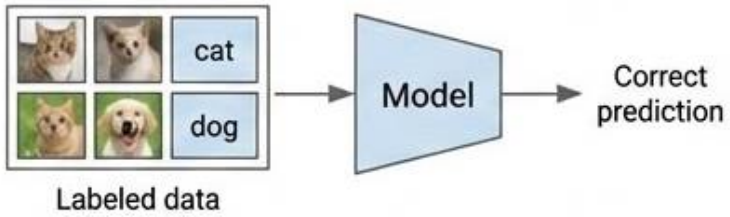
Performance Measure (P): A quantitative metric used to evaluate how well the system performs the task.

Examples of Performance Measures (P) for Different Tasks (T)

Task (T)	Example	Performance Measure (P)
Classification	Identifying spam emails	Accuracy, Precision, Recall, F1-Score
Regression	Predicting house prices	Mean Squared Error (MSE), Mean Absolute Error (MAE)
Clustering	Grouping similar customers	Silhouette Score, Davies-Bouldin Index

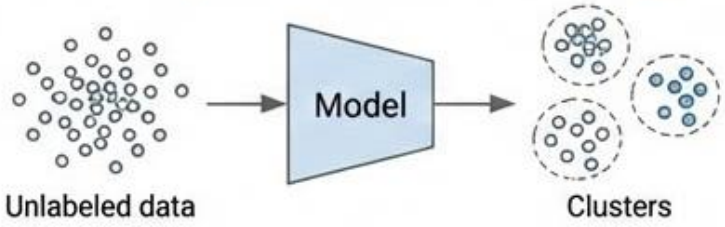
Types of Learning Experiences (E)

Supervised Learning



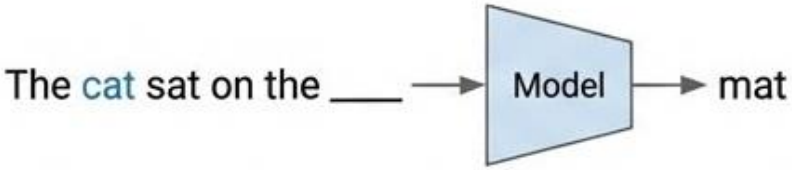
Learning from labeled data (input-output pairs). Goal is to map input to output.

Unsupervised Learning



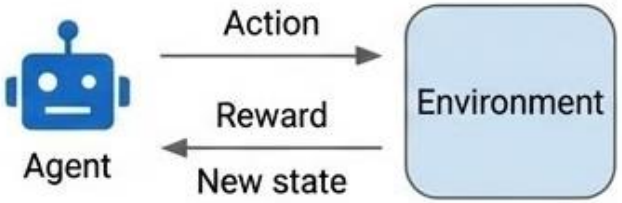
Learning patterns and structures from unlabeled data. No explicit target.

Self-Supervised Learning



Learning by generating labels from the data itself (e.g., predicting missing parts).

Reinforcement Learning



Learning through trial and error to maximize a reward signal over time.

From Machine Learning to Deep Learning: The Feature & Data Shift

Machine Learning (ML)

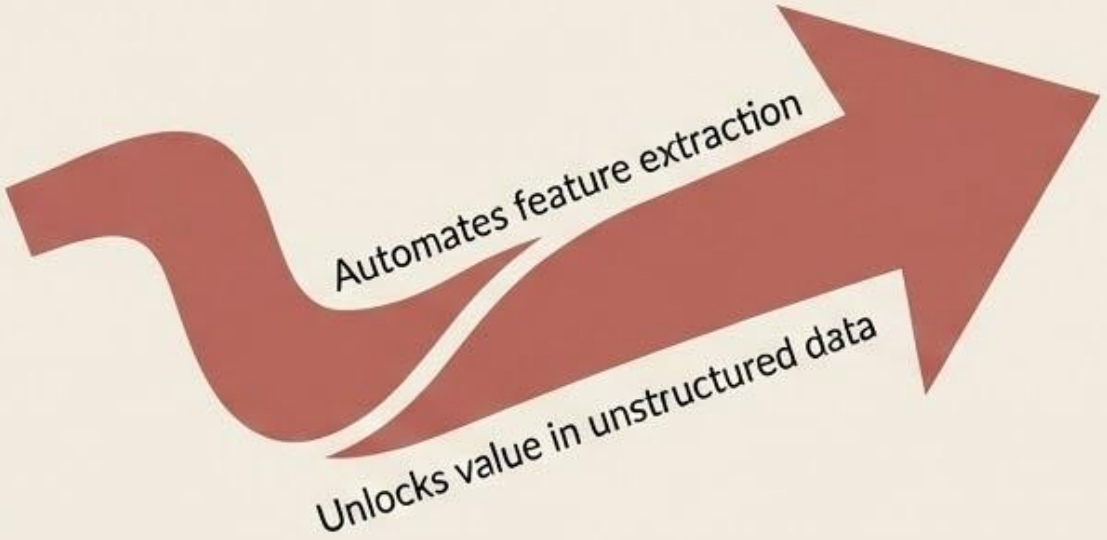
Feature Engineering

Manual process.
Domain expertise required to extract relevant features.
Time-consuming.

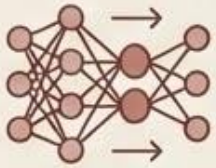
Data Preference

Structured Data.
Prefers organized, tabular formats (e.g., databases, CSVs).

Why the Shift?



Deep Learning (DL)



Feature Learning

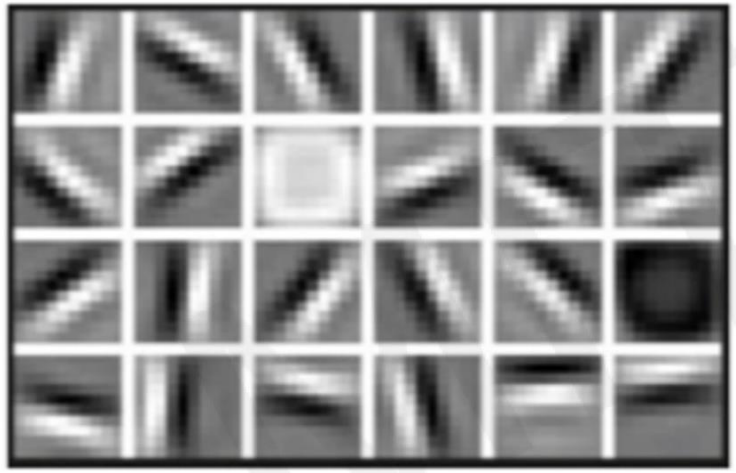
Automatic.
Model learns hierarchical features directly from raw data.
Evolving representations.

Data Preference

Unstructured Data.
Excels at finding patterns in complex, raw formats (e.g., images, audio, text).

Example

Low Level Features



Lines & Edges

Mid Level Features

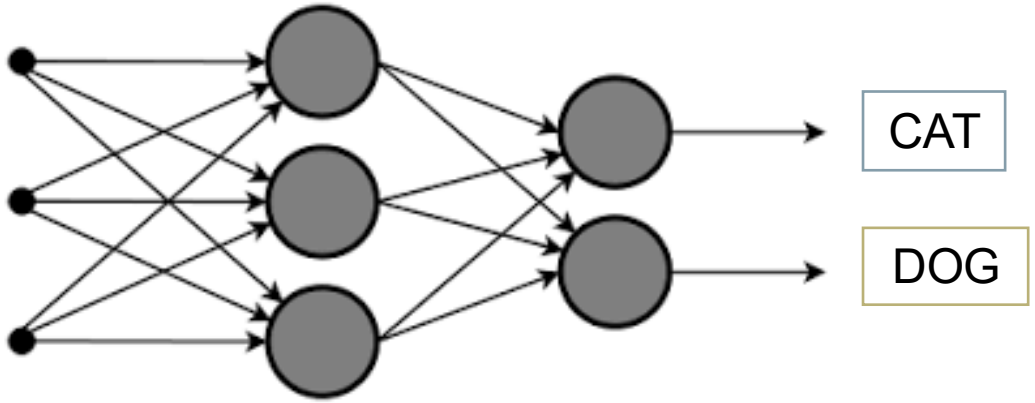
Eyes & Nose & Ears

High Level Features

Facial Structure

Example

Initial success of deep learning models were in classification tasks like the following.



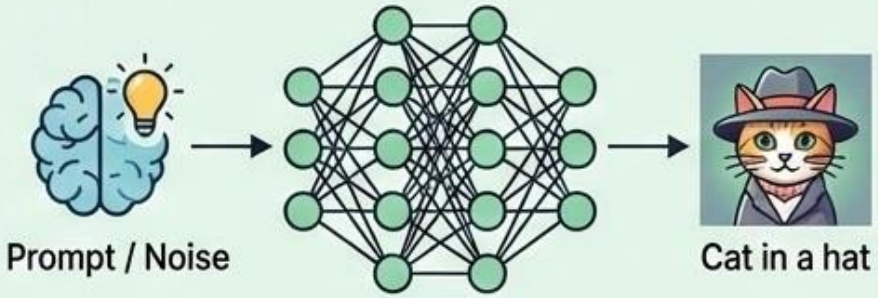
The Evolution of Deep Learning: From Classification to Generation

Initial Success: Image Classification

- Models trained to identify and categorize objects within images.
- Focus on discriminating between pre-defined classes.
- **Example:** Is this an image of a cat or a dog?

Evolution & Specialization

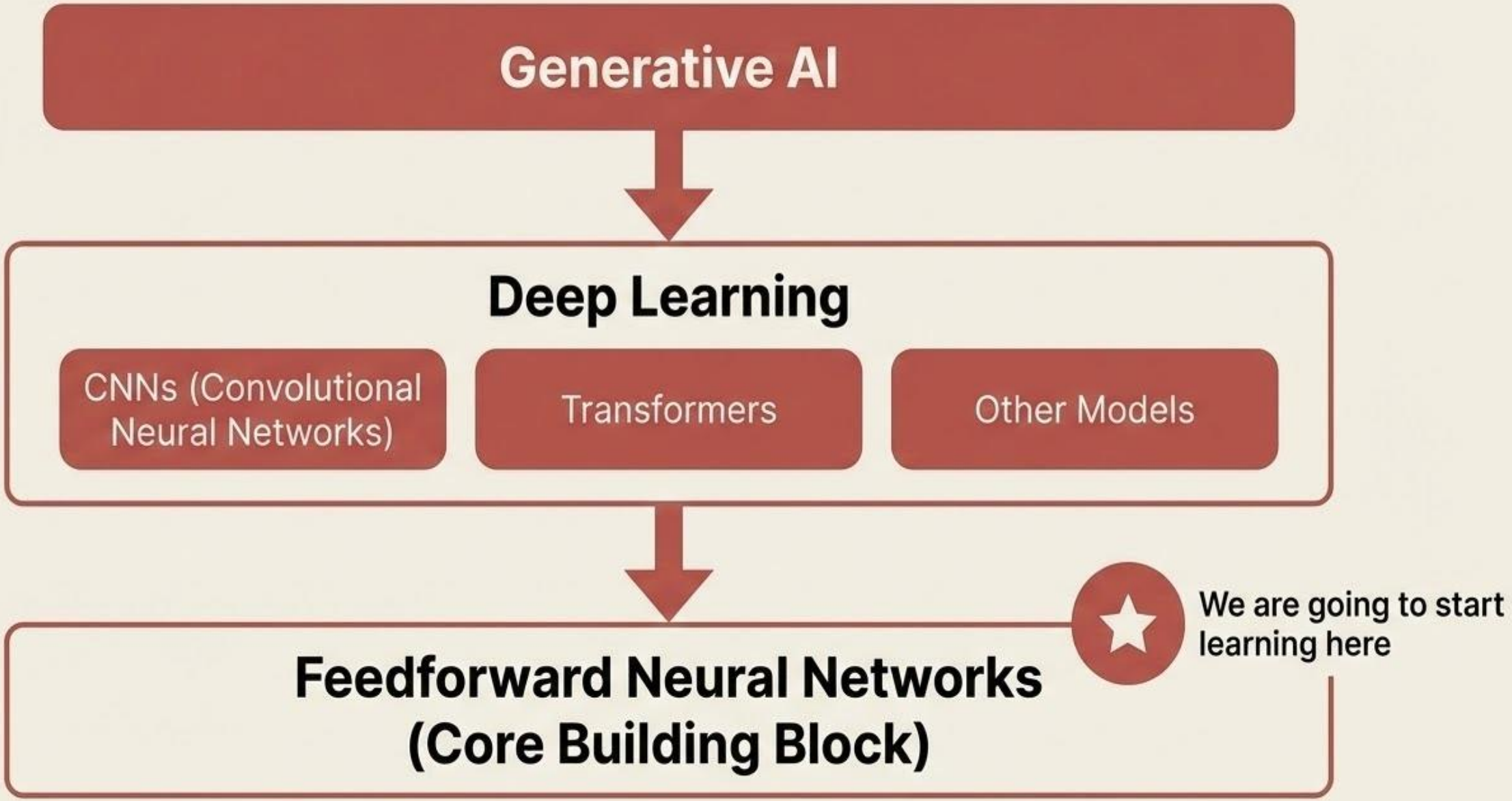
The Shift to Generative Tasks



- Models that can create new, realistic content (images, text, audio).
- Focus on understanding and reproducing the underlying data distribution.
- **Example:** Generate an image of a cat wearing a hat.

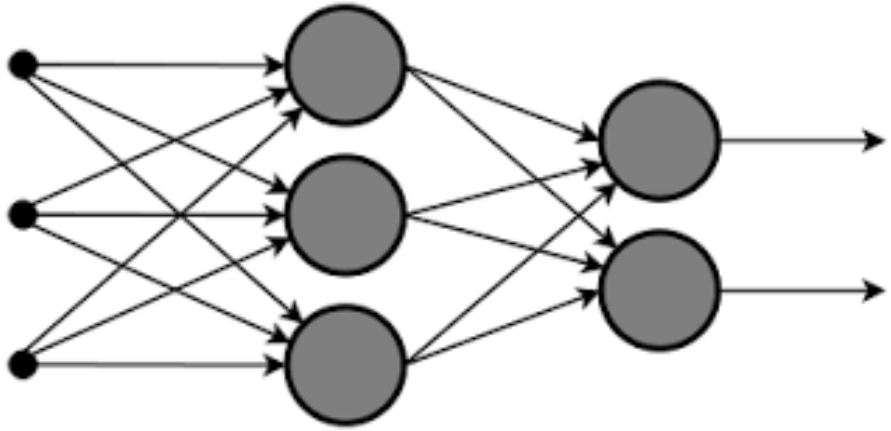
Deep learning has evolved from simply recognizing existing patterns to understanding them well enough to create new ones.

Generative AI explosion


Generative AI
Deep Learning
CNNs (Convolutional Neural Networks)
Transformers
Other Models
Feedforward Neural Networks (Core Building Block)
We are going to start learning here

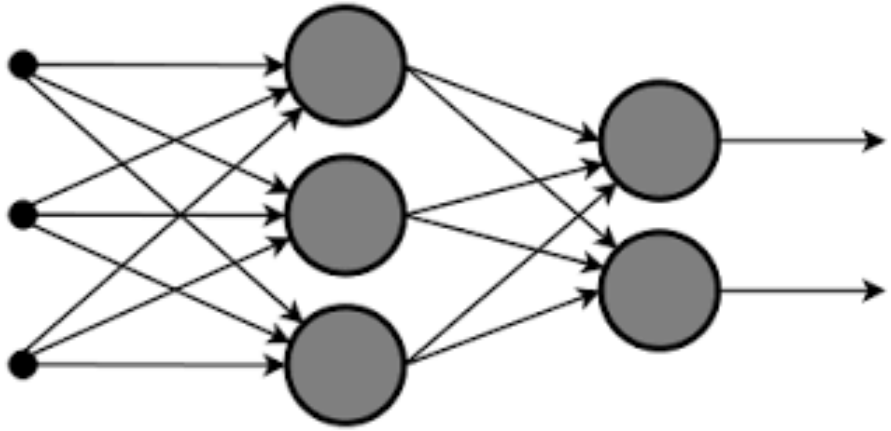
Feedforward Neural Network

- Feedforward neural network is a model built up from a series of interconnected layers.
- The output from one layer is the input for the following layer.



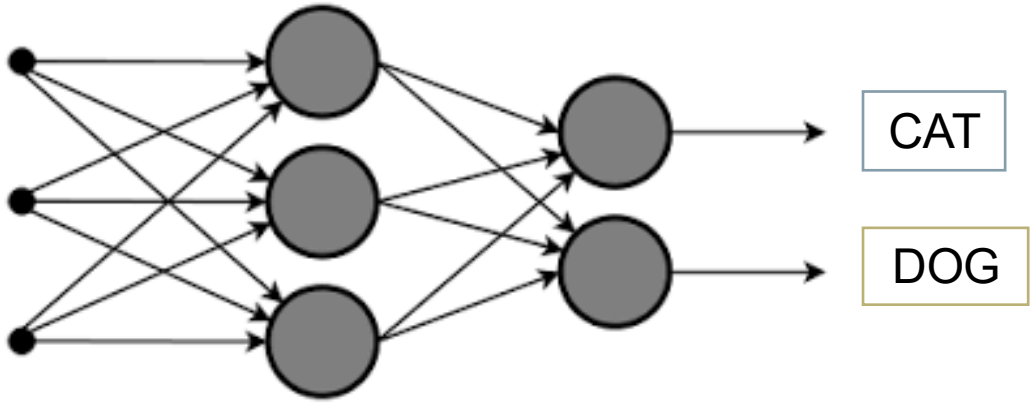
Feedforward Neural Network

- Each layer in the network is made up of individual nodes often called neurons.
- Neural networks are roughly motivated by human brains.



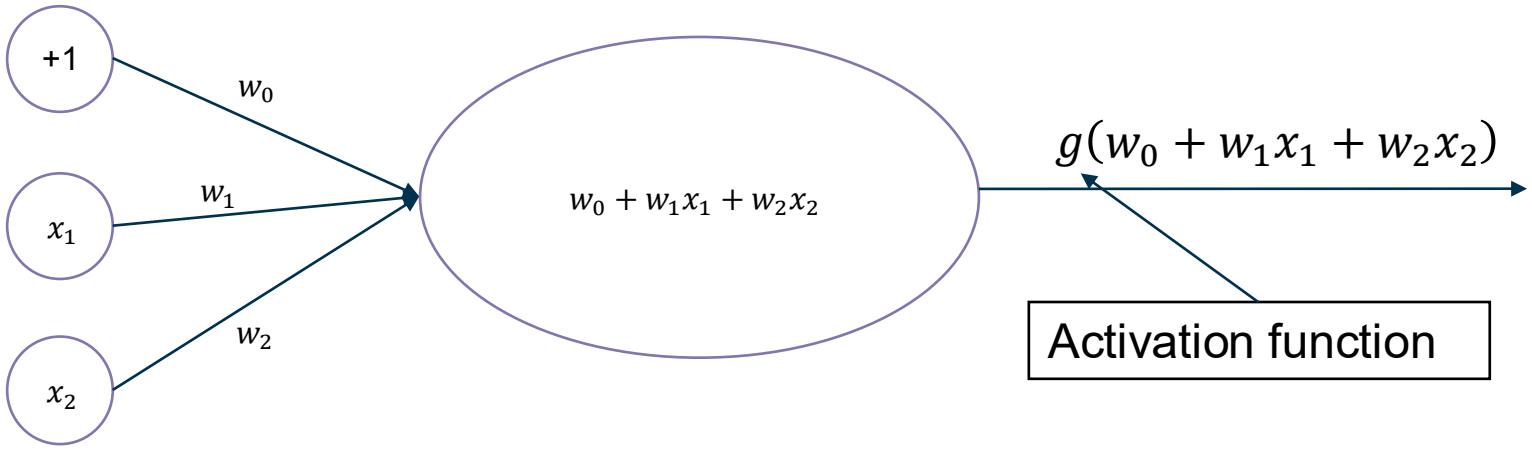
Example

We might want the network to tell us if an image is of a cat or a dog.

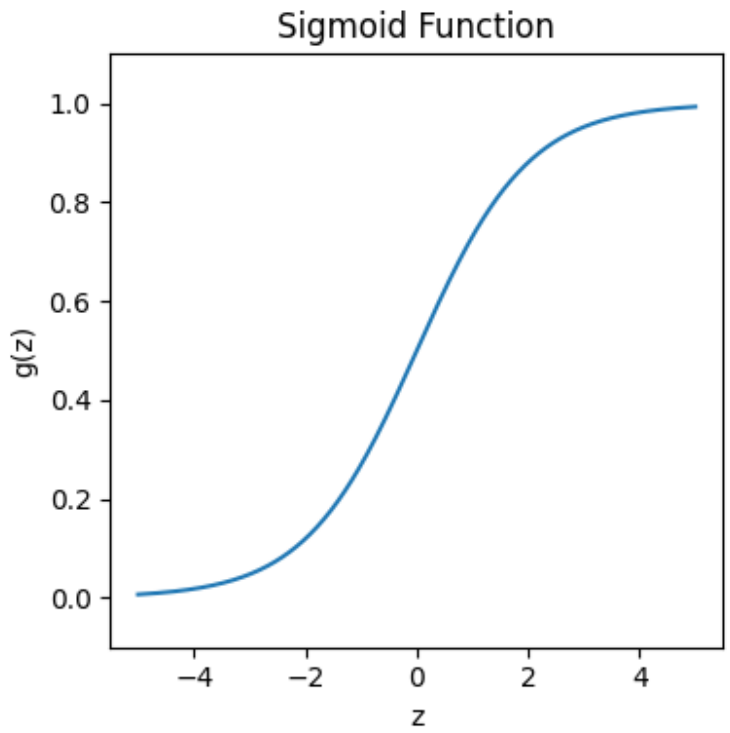


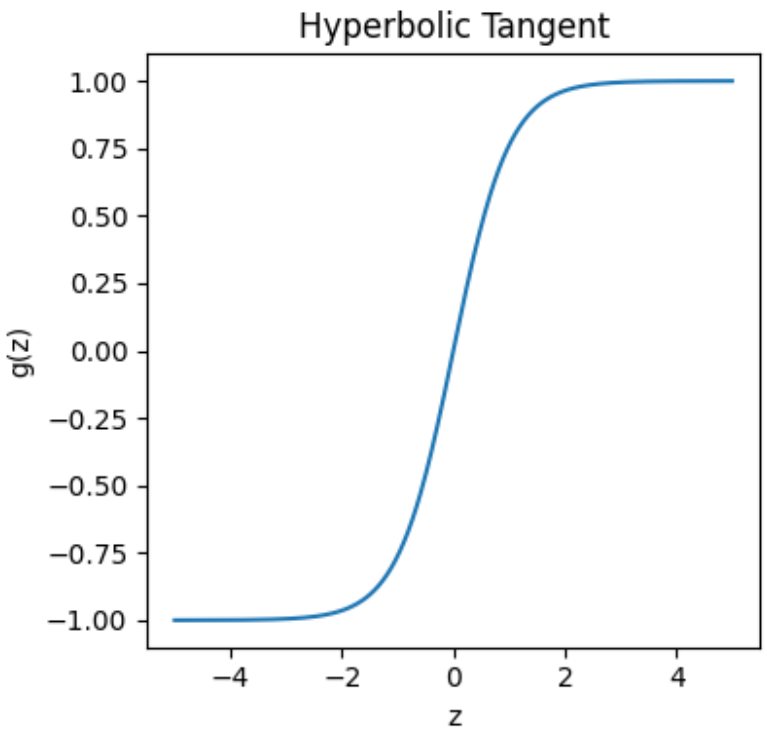
The Perceptron

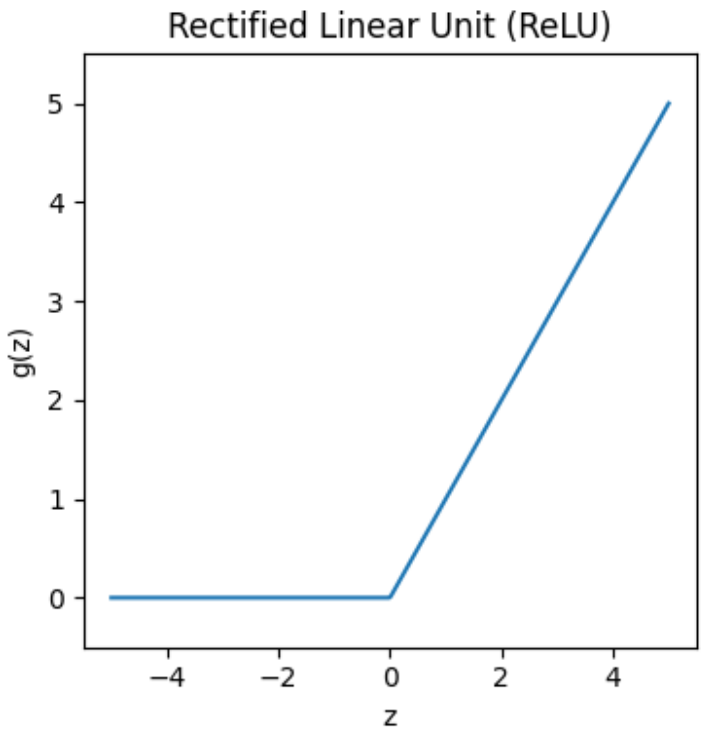
The basic building blocks of a neural network is the perceptron.



Common Activation Functions

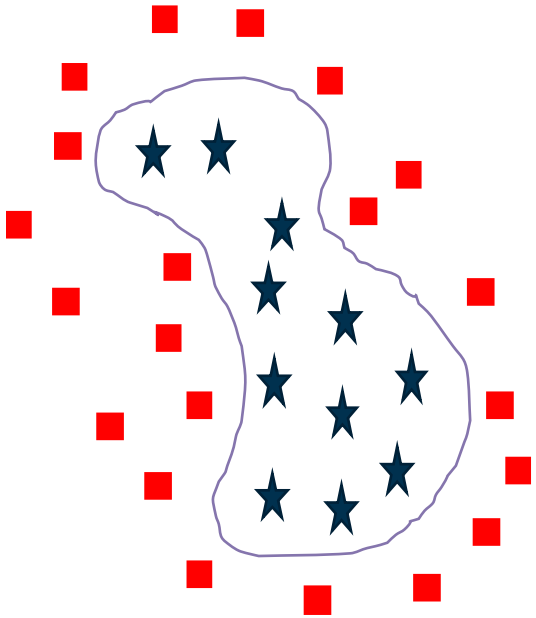






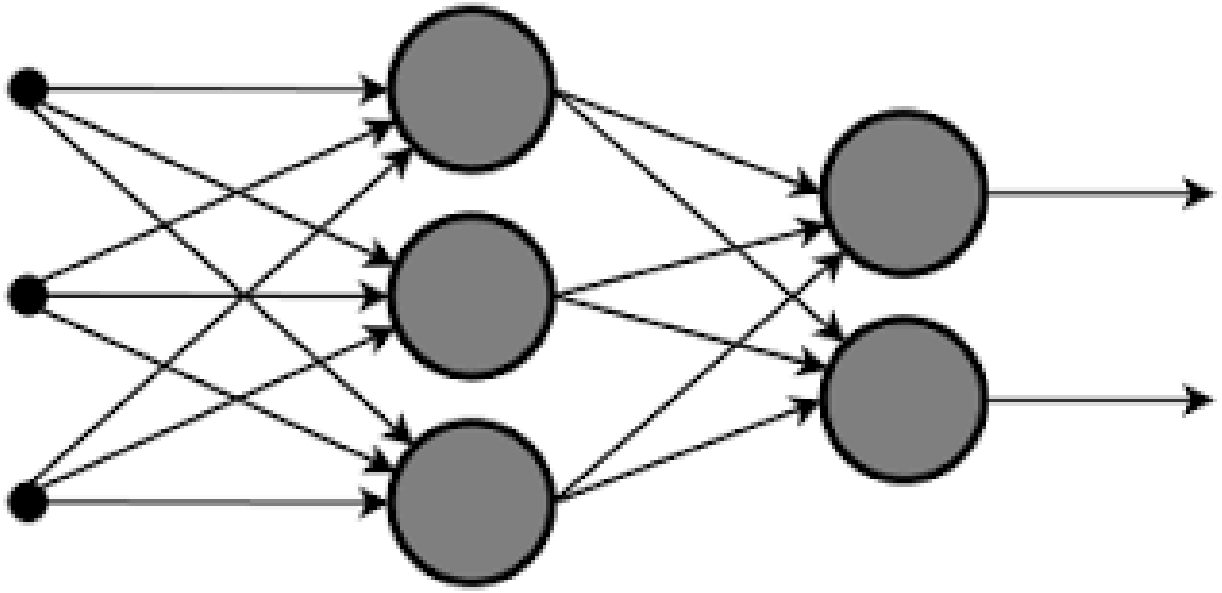
Comments

- Activation functions are non-linear.
- This enables more complex relationships to be represented.
- For instance, the following cannot be represented with linear activation functions.



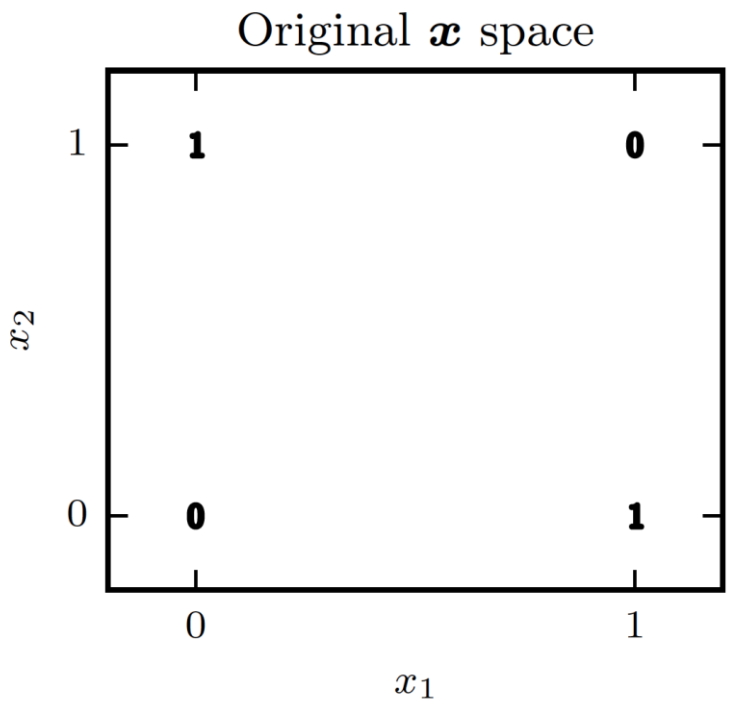
Neural Networks

- Neural networks can then be built up by connecting many perceptrons.



Example

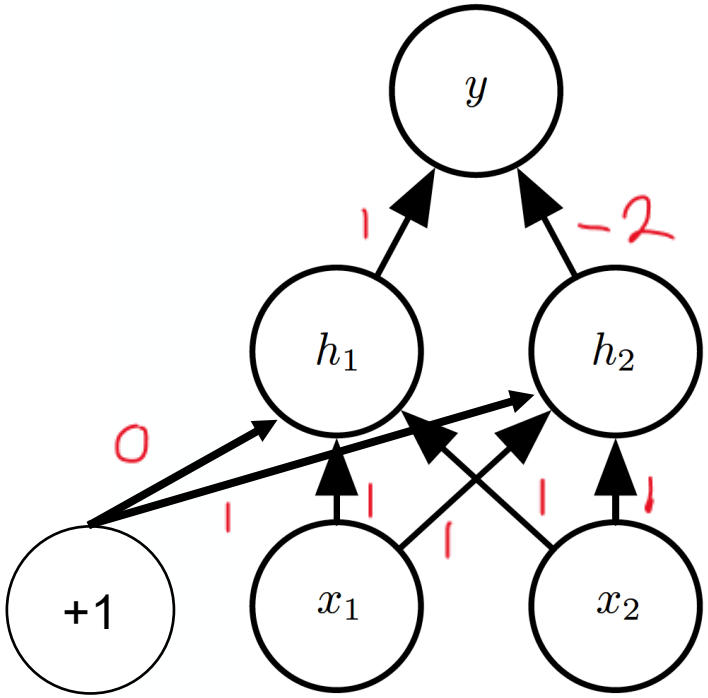
We are now going to follow an example from [1] Chapter 6. Consider the following data



Images taken from [1] Chapter 6.

Example

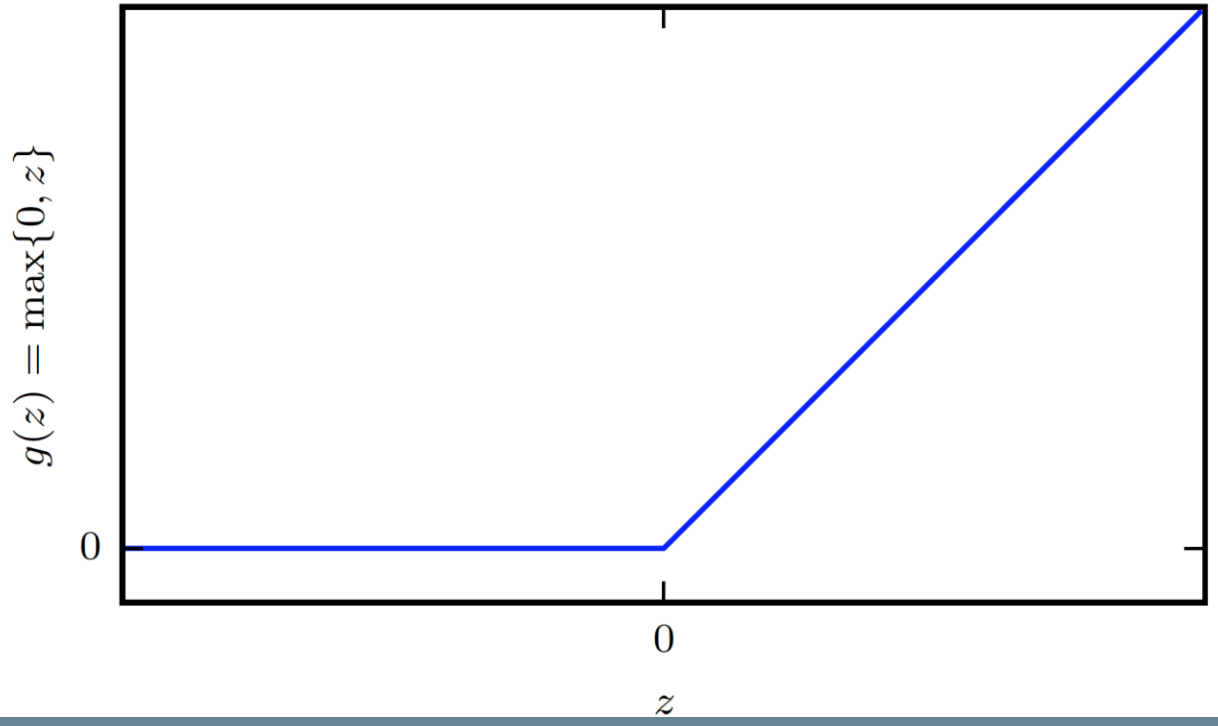
We have all the data and simply want to create a neural network which can execute the XOR operator perfectly.



Images taken from [1] Chapter 6.

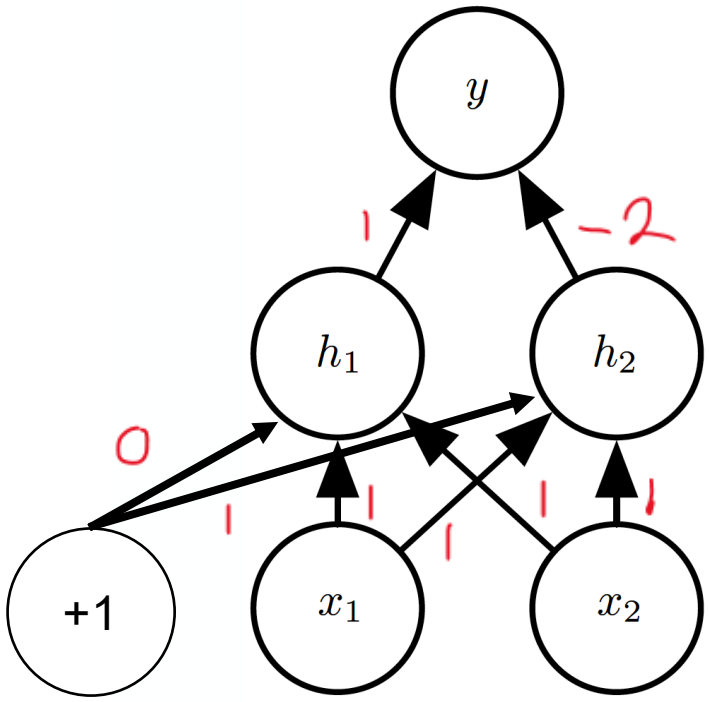
Example

As with any NN we need to choose an activation function for the neuron. A common one is ReLU:



Images taken from [1] Chapter 6.

Example



Images taken from [1] Chapter 6.

Suppose the inputs are (1,0) then the output should be 1.

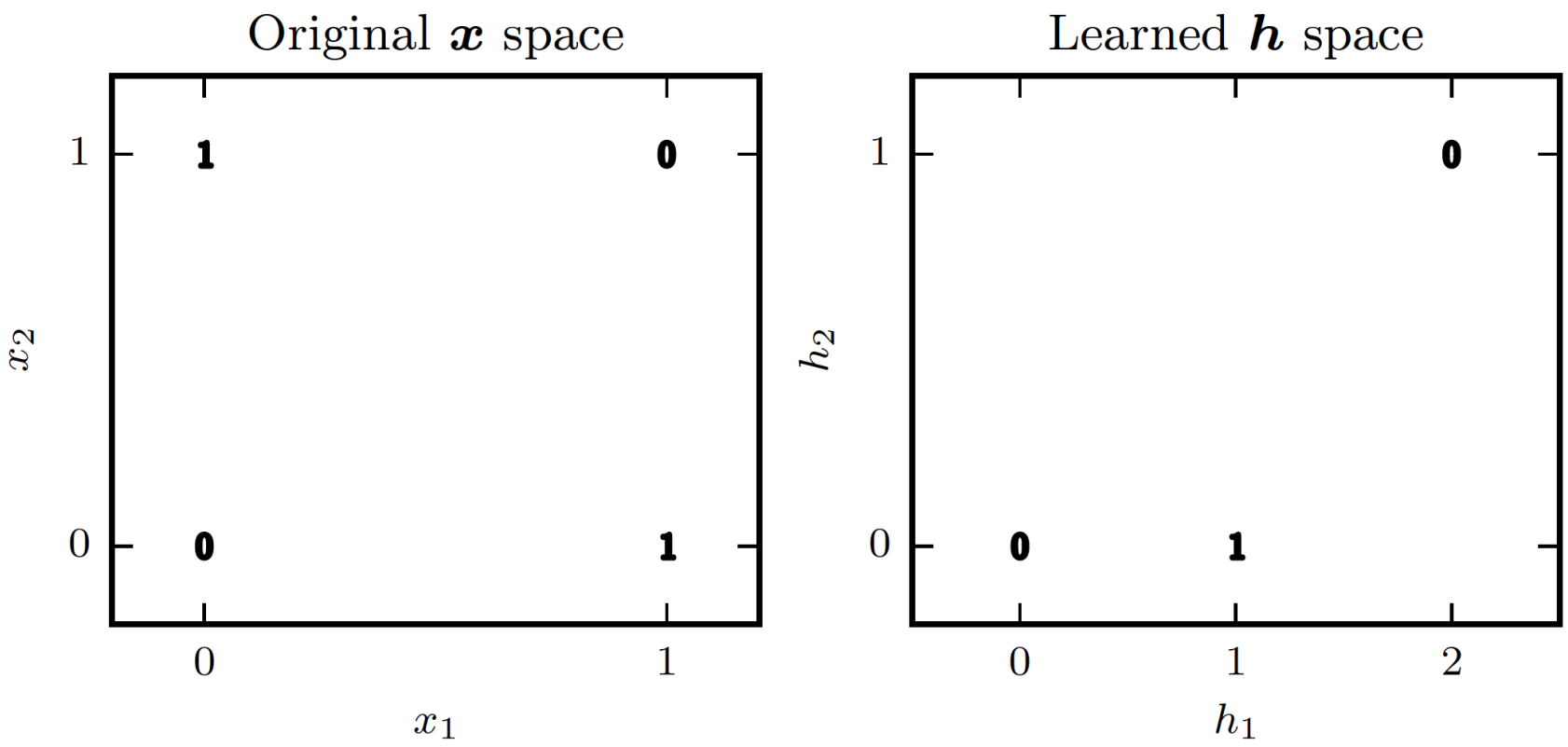
Let's check the output is correct.

Exercise

Verify that the network gives the correct answer for the other three inputs:

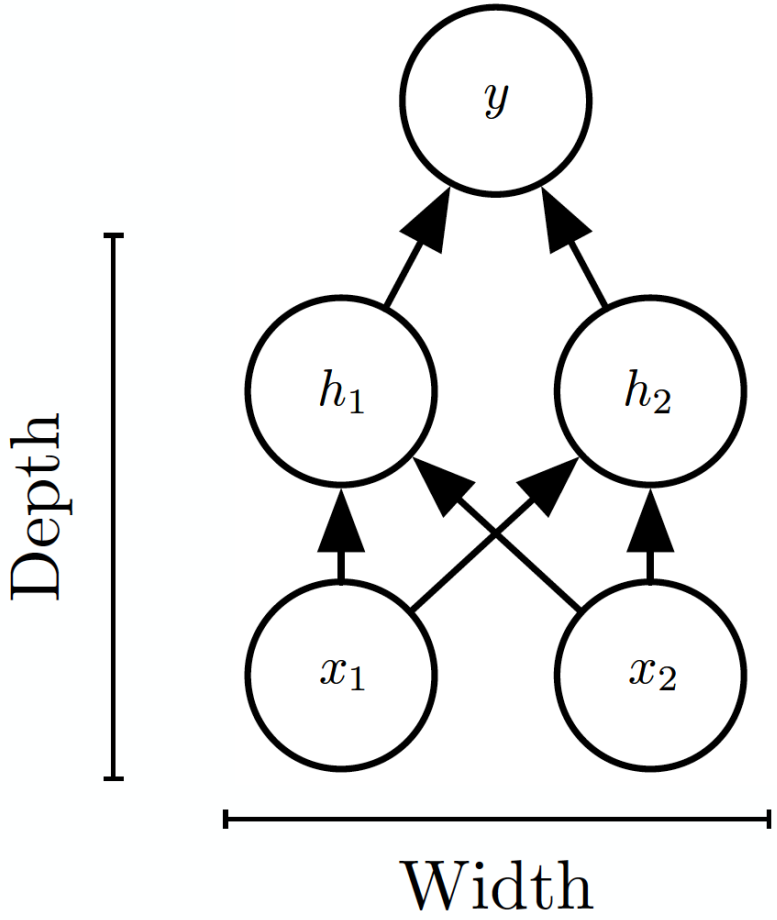
1. $x = (0,0)$
2. $x = (0,1)$
3. $x = (1,1)$

Example



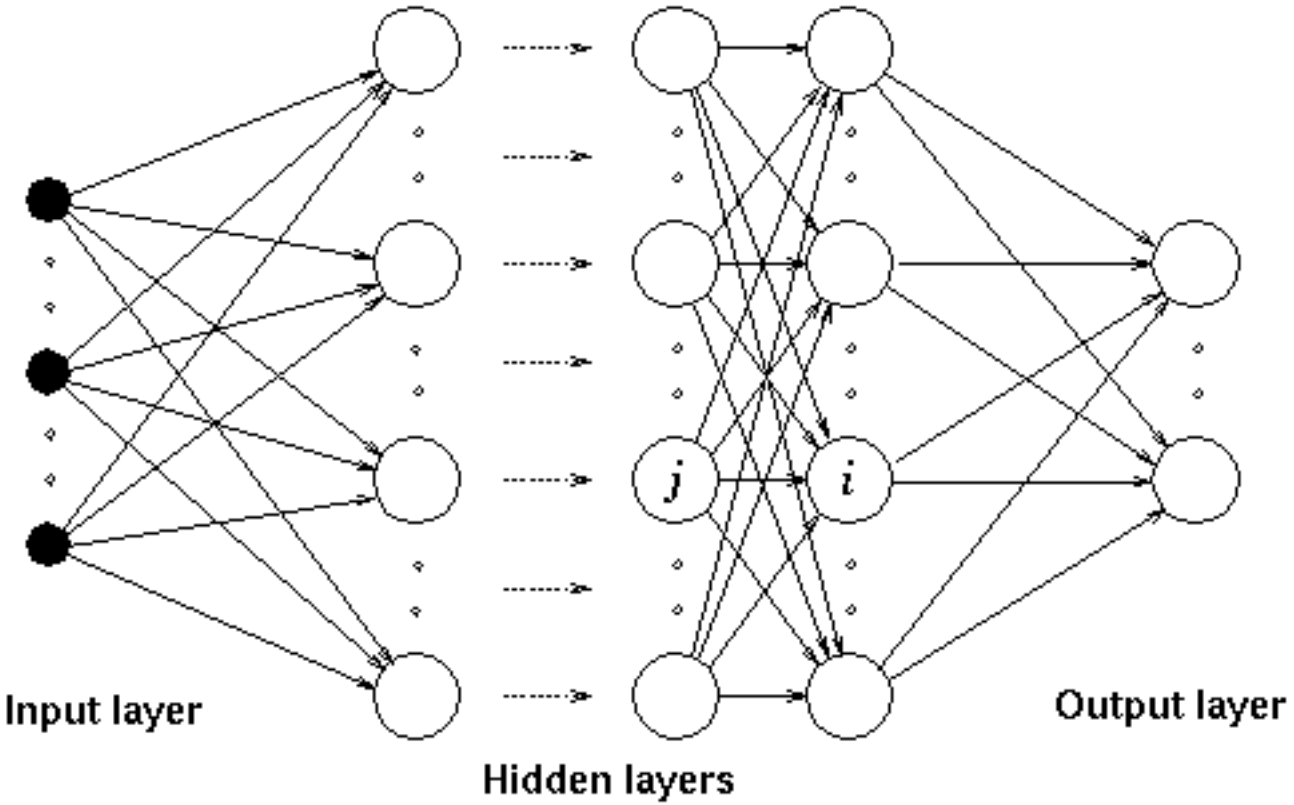
Images taken from [1] Chapter 6.

Network Architectures

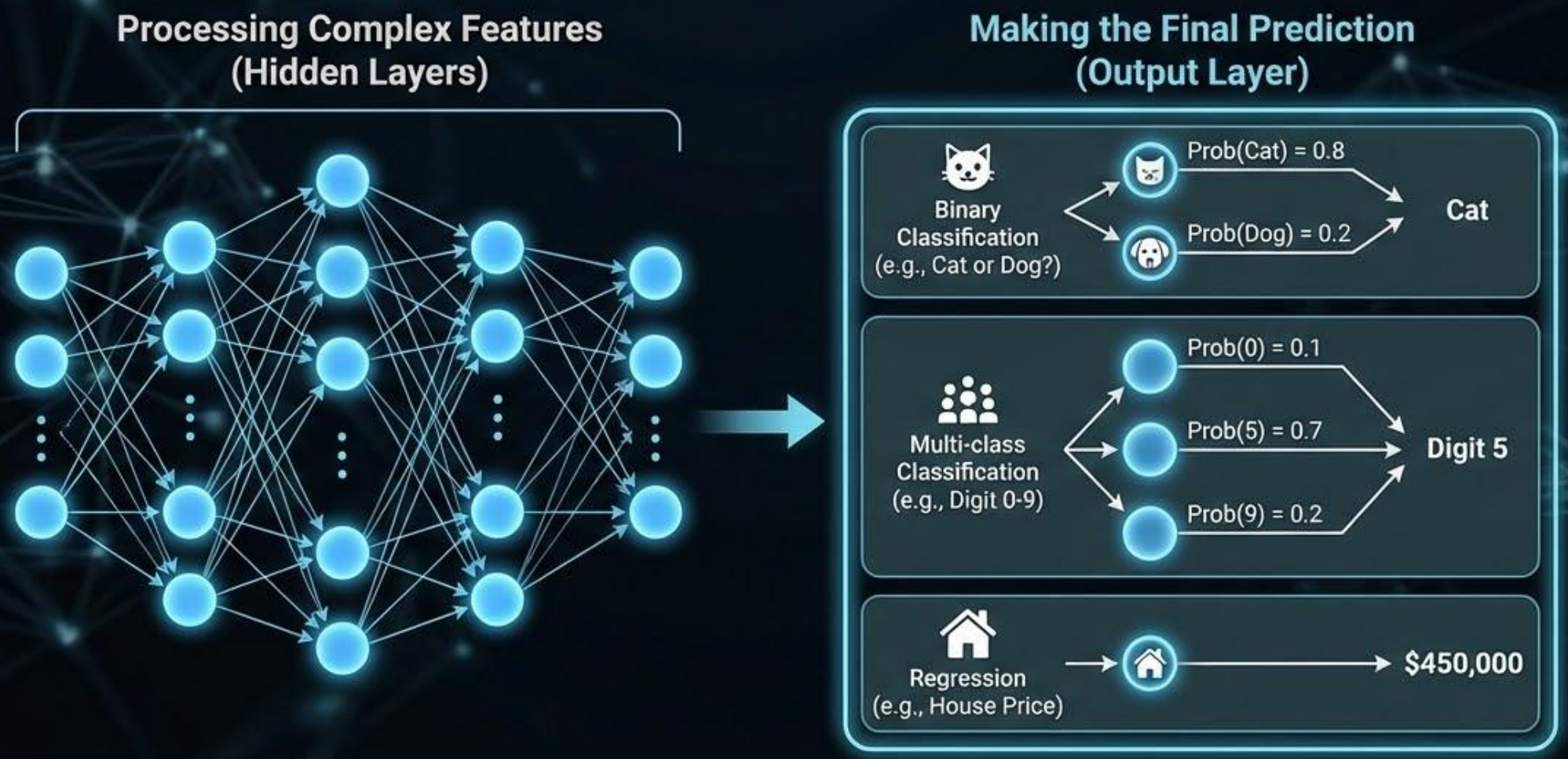


Images taken from [1] Chapter 6.

General Feedforward



The Output Layer: The Network's Final Decision



The output layer's structure and activation function determine how the network's internal representation is translated into a final, interpretable result for the specific task.

Connecting Output Layers to Cost Functions

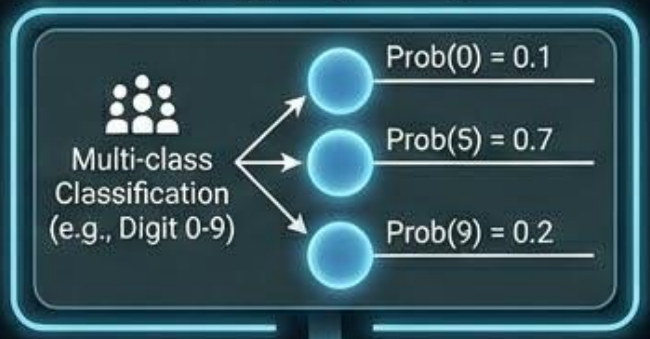
Binary Classification
(e.g., Cat/Dog)

Binary Cross-Entropy

Penalizes confident wrong predictions. Measures difference between two probability distributions.

$$- (y * \log(p) + (1-y) * \log(1-p))$$

Multi-class Classification
(e.g., Digit 0-9)



Categorical Cross-Entropy

Measures the difference between the predicted probability distribution and the true distribution.

$$- \sum (y_i * \log(p_i))$$

Regression
(e.g., House Price)

Mean Squared Error (MSE)

Calculates the average of the squares of the errors (difference between predicted and actual values).

$$\frac{1}{n} * \sum (y_i - \hat{y}_i)^2$$

Output Layer & Cost Function

- Generally, the function used for the output layer will correspond to as previously mentioned.
- Further, the type of output layer will define the cost function we use to train the network.

y	g_o	Cost Function
Binary	Sigmoid	negative log-likelihood
Discrete	Softmax	negative log-likelihood
Continuous	Identity	Mean Squared Error

Good to know

For any *reasonable function* we can *approximate* it to *any degree of accuracy* using a *feedforward neural network with only one hidden layer*.

- The more accurate we want our approximation the wider we need to make our hidden layer.
- Networks with fewer layer tend to be more prone to overfitting.

The challenge of ML

The challenge at the heart of ML is our models need to perform well on new (previously unseen) data i.e. the models need to generalise to new data.

- Theoretically, we quantify how well a model generalises on new data to be the expected value of the performance measure on new data.
- In practice, we estimate this with the value of the performance measure on the test set.

Test & Training Sets

- Suppose we have a ML model that is suppose to perform some task T.
- Training Data: Data used to train the model. Often training is done by optimising the performance measure P on the training data.
- Test Data: We use this data to evaluate the performance of our model. Use the performance measure P on the test data to quantify performance.
- Test data should not be used in training of the model.

Splitting Data into Training & Test Sets

- Given some initial dataset for our model development we split it into training & test sets.
- Often we use 70/30 or 80/20 percentage splits.
- Important to randomly split your data according to these percentages as the might be some bias in the way data was recorded.

Overfitting & Underfitting

- A common problem in ML is that our model will perform well on the training data but generalise poorly i.e. perform badly on the test data.
- This is known as *overfitting.*
- Another problem that can occur is that our model performs badly on the training sets.
- This is known as *underfitting.*

References

1. *Deep Learning* by I. Goodfellow, Y. Bengio and A. Courville
2. R. Ramakrishnan and MIT Sloan School of Management, *15.773 (S24): Lecture 01: Introduction to Neural Networks and Deep Learning*. MIT OpenCourseWare, Massachusetts Institute of Technology, 2024. [Online]. Available: https://ocw.mit.edu/courses/15-773-hands-on-deep-learning-spring-2024/mit15_773_s24_lec01.pdf
3. Google DeepMind, “Gemini Image (Nano Banana),” Google DeepMind, 2025. [Online].